Hot Air, Cold Body

3A — 27 October 2026

Lab sheet: how long ago did the potato leave the bath?

Set up → **KEY 1 Check zero** → **Potato in** → **KEY 2 Record 20 min** → **Graph + gradient** → **KEY 3 Extend back** → **Report**

You need

- Arduino and circuit from Lab 1, laptop
- `hotair_cooling_logger.ino`, Arduino IDE
- your calibration line from Lab 1: $N = N_0 + S\,T$
- ice, water, graph paper (mm), calculator, the report file (Word)
- your potato (in foil) and its tag

You hand in (one per group, before the end of the lesson)

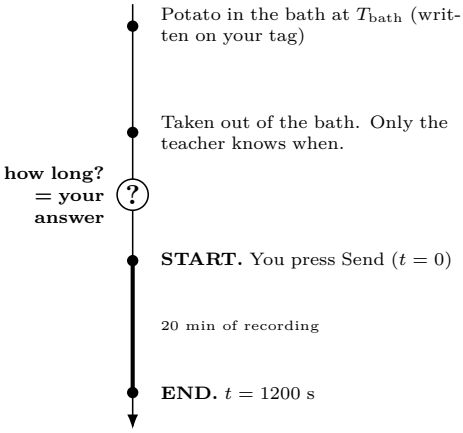
- the **graph**, on paper
- the **report** (Word file). You answer it **last**, after the graph. The data table is inside it.

LEVEL 1 The question

Case file. A potato was taken out of a hot bath at a time that only the teacher knows. You will watch it cool for 20 minutes.
Your question: how long before you pressed Start was it taken out of the bath?

The idea in 3 lines

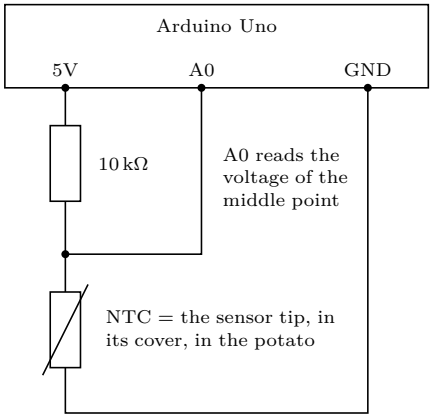
- A hot potato cools like this: $T - T_{\text{room}} = (T_0 - T_{\text{room}})\,e^{-kt}$.
- Take the logarithm: $\ln(T - T_{\text{room}}) = \ln(T_0 - T_{\text{room}}) - k\,t$. On a graph of $\ln(T - T_{\text{room}})$ against t this is a **straight line with gradient $-k$**.
- Go back along the line until you reach the height of the bath, $\ln(T_{\text{bath}} - T_{\text{room}})$: that is the moment the potato left the bath.



LEVEL 2 How to get the data

1 Circuit

- Build the *same* circuit as Lab 1: same NTC, same 10 kΩ resistor, same pin A0.
- Change nothing: your calibration belongs to this sensor + resistor + pin.
- The sensor tip stays inside its waterproof cover (freezer bag or nitrile finger).



2 Sketch

- Open `hotair_cooling_logger.ino`.
- At the top type *your* N_0 and S from Lab 1.
- Upload it (Upload arrow, top left).

3 Serial Monitor

- Tools → Serial Monitor. Set **9600 baud** (bottom right).
- One `live` line appears every second. Use it for steps 4–6.
- To stop or restart: press Reset on the Arduino.

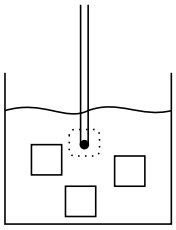
```
live  N=512.0  T=23.40 C
live  N=512.0  T=23.38 C

live  N=511.0  T=23.45 C
```

KEY MOMENT 1 — Is the thermometer still right?

4 Ice point (1 min)

- Sensor (in its cover) in ice + a little water. Stir gently, do not touch the bottom or the wall.
- After 1 minute read the live T: it should be $0 \pm 1\,^{\circ}\mathrm{C}$.
- Report, Part A: write the difference from 0 as your **zero error**.
- More than $1\,^{\circ}\mathrm{C}$ off? Use the checklist before you go on.



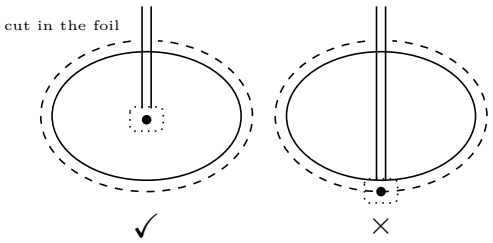
ice + a little water; sensor tip in its cover. Target: $0 \pm 1\,^{\circ}\mathrm{C}$

5 Room temperature (1 min)

- Sensor in the air, away from hands, laptop and sun.
- After 1 minute write 5 live readings and their mean T_{room} (Report, Part A).

6 Your potato

- Take your potato (hot, in foil) and its tag from the teacher.
- Write T_{bath} from the tag in the report (Part A). With the calculator find $\ln(T_{\mathrm{bath}} - T_{\mathrm{room}})$ and write it too: it is the height of your bath line.
- Push the covered sensor through the small cut in the foil to the **centre**. Do not push it out of the other side.
- Wait 1 minute: the live reading falls slowly but must stop jumping.



KEY MOMENT 2 — Record the cooling curve (20 min)

7 Start

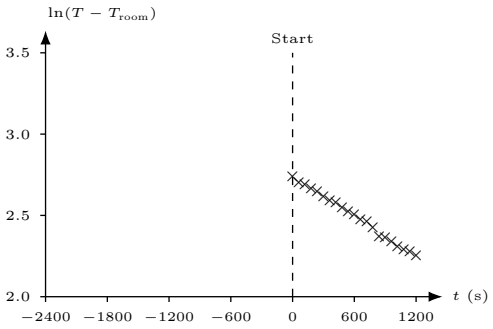
- Write down the clock time (Report, Part A).
- In the Serial Monitor type one letter and press **Send**: the run starts, $t = 0$.
- A line t_s, N, T_C appears, then one row every 10 s for 20 minutes.
- **Do not touch the potato, the sensor or the foil.**

```
t_s   N       T_C
0     370.1   37.45  <--
10    372.3   37.25
20    373.1   37.18
30    372.9   37.20
40    373.0   37.19
50    374.5   37.06
60    376.0   36.93  <--
...
```

What you see. Use the rows marked <--: one per minute, $t = 0, 60, \ldots, 1200$ s.

8 While it records: table and graph paper

- Axes as in the figure: $\ln(T - T_{\mathrm{room}})$ upwards, t in s to the right, from -2400 s to $+1200$ s. Line $t = 0$: write Start on it.
- Vertical scale: the points must fill more than half of the page **and** $\ln(T_{\mathrm{bath}} - T_{\mathrm{room}})$ must also fit.
- Every 60 s (rows marked <--): write T in the report table, calculate $T - T_{\mathrm{room}}$ and its ln, mark the point $\times$.



Invented data, to show what your paper should look like.

9 When you read DONE

- The run stops by itself. Select all the rows from `t_s` to the last one, copy, paste into Google Sheets or Excel (the columns split by themselves) and save: it is your backup.
- Check your 21 report rows against the backup.
- Clean up: foil and covers in the bin, potato to the teacher (do not eat it), ice to the sink, sensor dry, circuit left connected.

If it does not work: check in this order, before you call the teacher

What you see	Check first	Then check
Nothing on screen, or strange symbols	Serial Monitor at 9600 baud; the right port selected (Tools → Port)	Unplug and replug the USB cable
"Port busy" or the Serial Monitor will not open	Close any other Arduino window and any Python script	Unplug and replug the USB cable, reopen the Serial Monitor
N is always about 0 or about 1023	NTC and resistor are both in the circuit; the A0 wire is pushed in	A loose wire: touch each one
N does not change when you warm the sensor with your fingers	The A0 wire is on the *middle point* between resistor and NTC	Is it the 10 kΩ resistor (brown–black–orange)?
N looks sensible but T is nonsense (hundreds of degrees)	N_0 and S typed in with the right sign, sketch uploaded again	Same sensor, resistor and pin as in Lab 1?
Ice point is more than 1 °C off	Cover is waterproof; ice and water mixed; sensor in for a full minute	Write it as zero error: it is data, not a failure

Optional, for experts: automatic logger (not required). Once: `py -m pip install pyserial`. Close the Serial Monitor, run `py hotair_logger.py` in its folder: it asks T_{room} and T_{bath} and saves `hotair_table_….csv` (21 rows with $T - T_{\text{room}}$ and ln). You still draw the graph by hand.

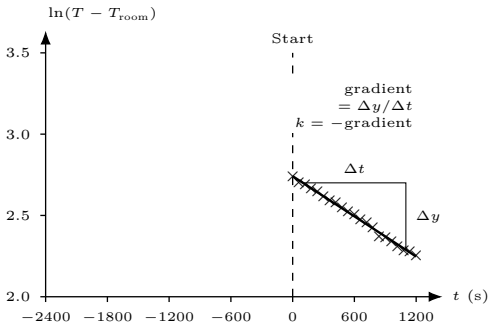
LEVEL 3 After the recording: graph and report

10 Best-fit line

- Draw a **straight line of best fit** through your points (about the same number above and below).
- **Do not** just join the first and the last point.

11 Gradient

- Draw a gradient triangle using more than half of the line.
- Gradient $= \Delta y/\Delta t$ (it is negative). Then $k = -$gradient, in s^{-1}.
- Report, Part C: gradient and k.



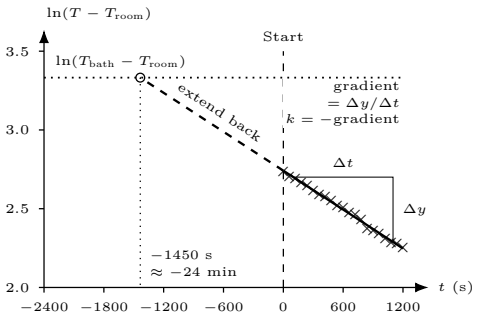
KEY MOMENT 3 — Extend the line back to the bath

12 The bath line

- Draw a horizontal dotted line at $\ln(T_{\text{bath}} - T_{\text{room}})$ (the number you calculated in step 6).
- Extend your best-fit line to the **left**, dashed, until it meets the dotted line.

13 Read the time

- Read t at the meeting point. It is negative: $-X$ s.
- $X/60$ is the number of minutes before Start at which the potato left the bath.
- Clock time of Start $- X =$ time it left the bath. The teacher will tell you the true time.



14 Report

- Now, and only now, open the report file and answer Parts A–D.
- Hand in the report and your graph before the end of the lesson.

LEVEL 4 Bonus

Bonus points, only if everything above and the whole report are complete.

1. **Half-life of the difference.** The time for $T - T_{\text{room}}$ to halve is $t_{1/2} = \ln 2/k$: calculate it.
2. **Why does the start bend?** Heat travels from the centre to the surface: what happens at each in the first minutes?
3. **Why a potato?** Is the k of a potato valid for a human body? What changes k?